



SVEUČILIŠTE U ZAGREBU

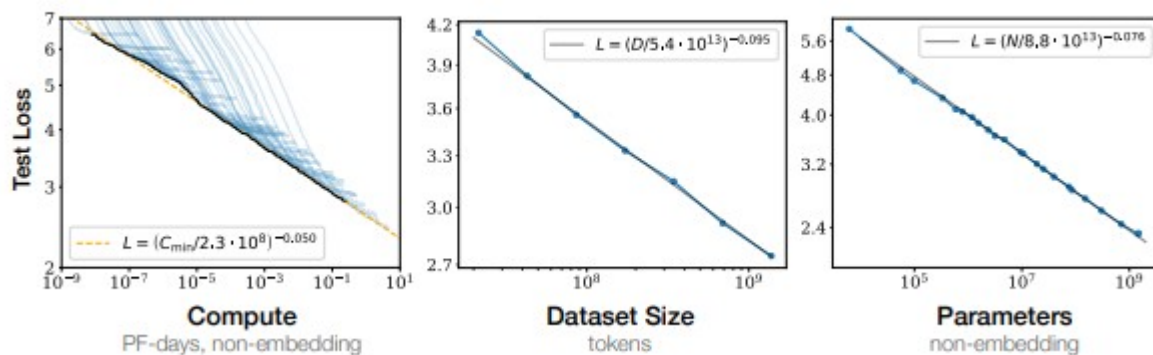
Fakultet
elektrotehnike i
računarstva

Prezentacijski seminar

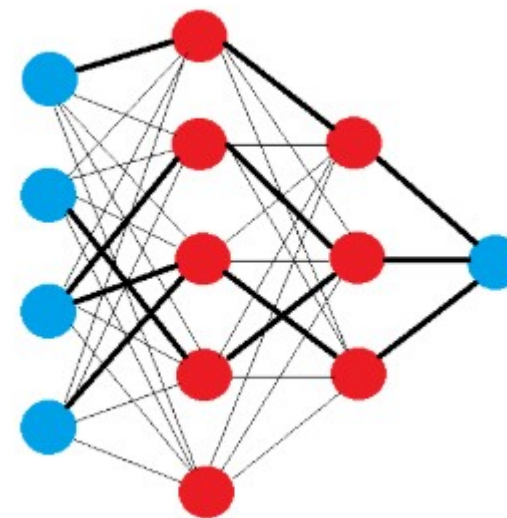
Evolucijska optimizacija kompresije transformerskih modela

Tonka Šegvić

- Zakon skaliranja [4]



- Redundancija
- “Lottery ticket hypothesis”



Orezivanje (pruning)

- **Strukturirano**

- uklanja cijele fizičke komponente, poput glava pažnje ili čitavih transformatorskih blokova
- ubrzanje na standardnim GPU

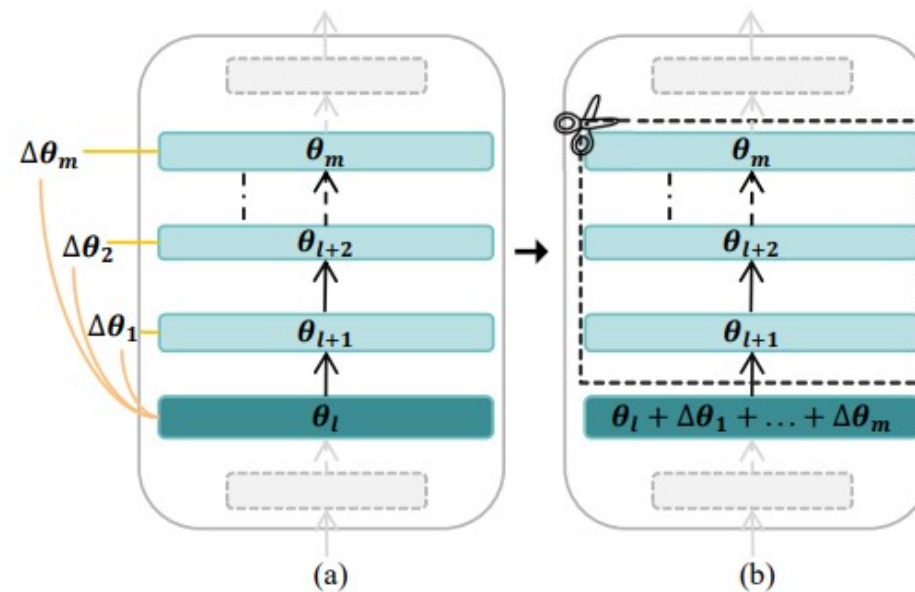
- **Nestrukturirano**

- uklanja pojedinačne parametre
- ne donosi stvarno ubrzanje na standardnim GPU bez specijalizirane hardverske podrške

Strukturirano orezivanje

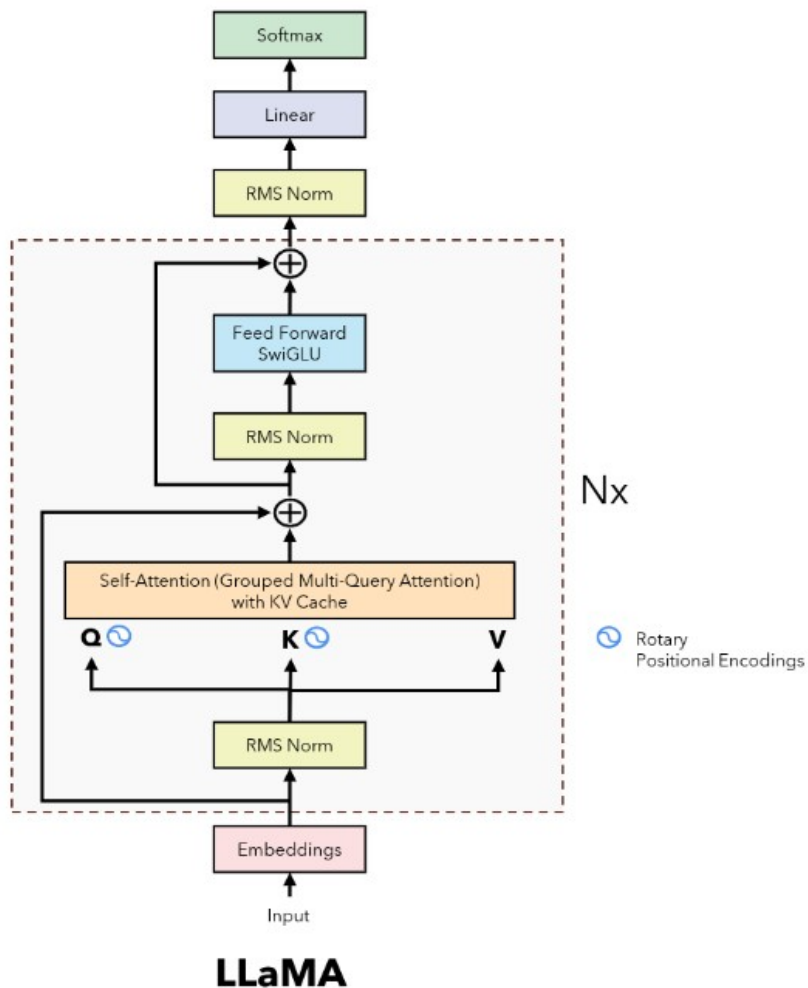
- ShortGPT [1]
 - Kosinusova sličnost između ulaza i izlaza u transformerski blok
 - Uklanjanje čitavih slojeva
- Diferencijalno spajanje parametara
 - LaCo [2]
 - “agregacija znanja”

$$\theta_l^* = \theta_l + \sum_{k=1}^m (\theta_{l+k} - \theta_l)$$

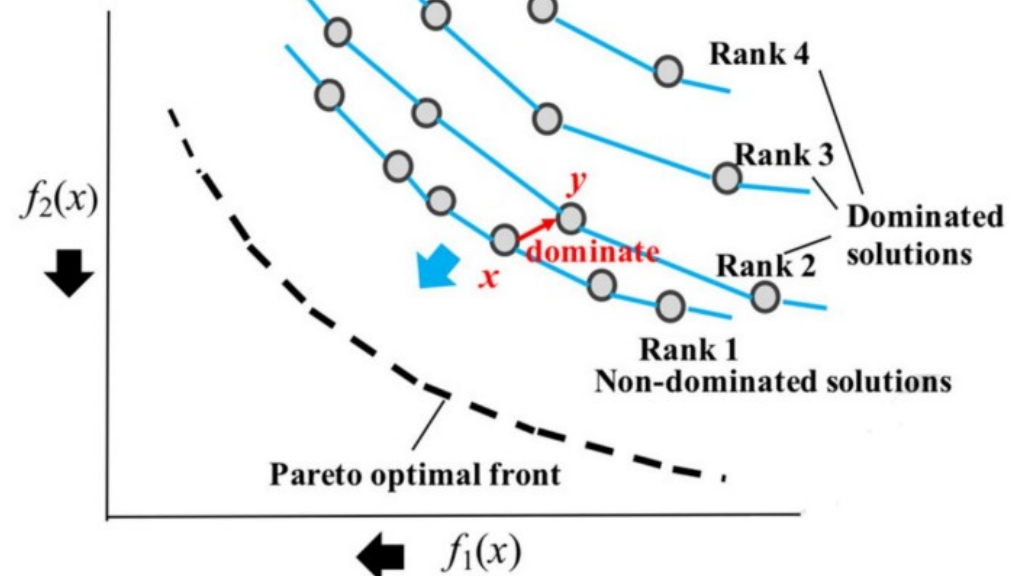
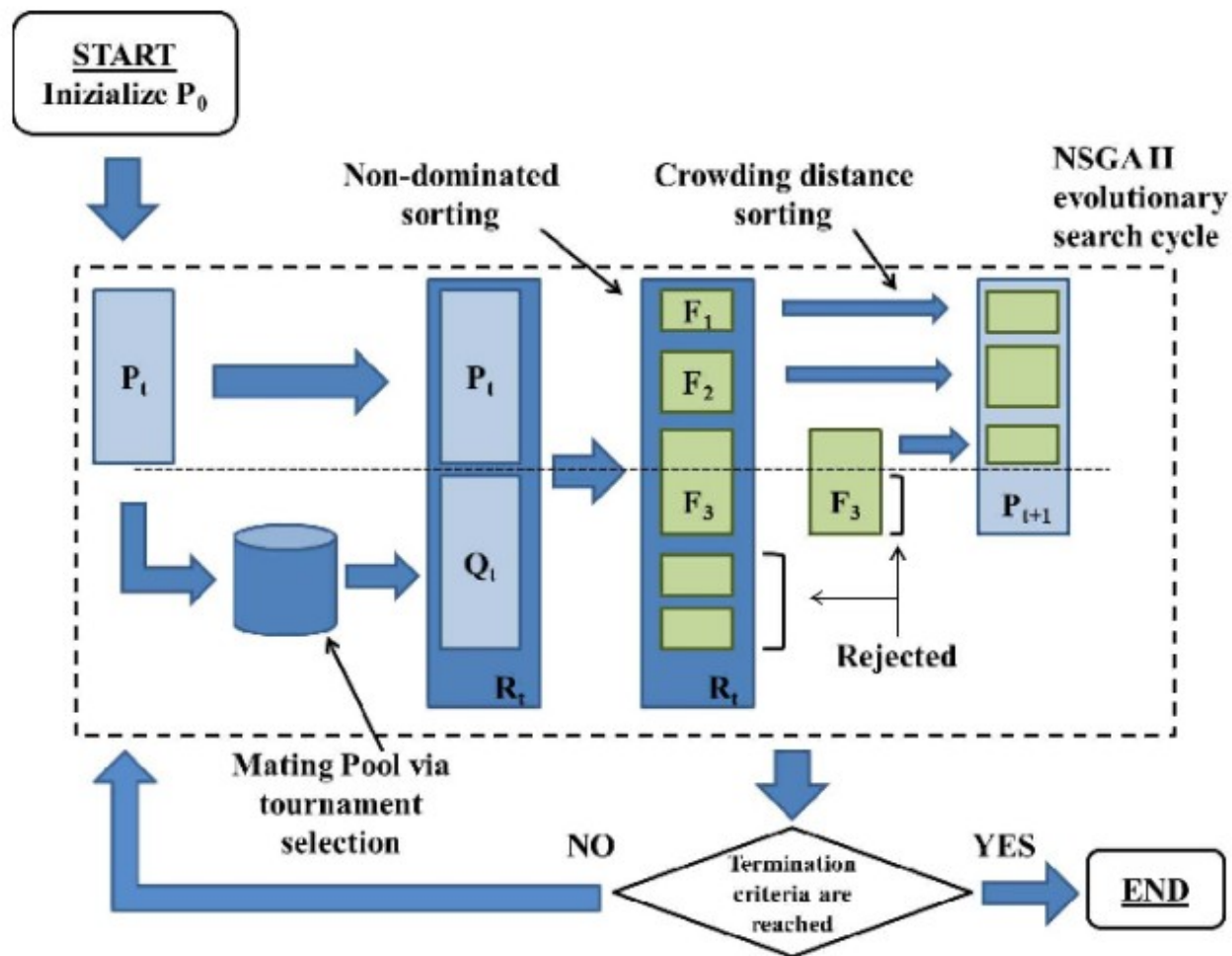


- Kako odabrati koje slojeve spojiti?
- Jednokriterijska optimizacija
 - Za fiksni omjer kompresije pronaći najbolje spajanje slojeva
- **Višekriterijska optimizacija**
 - Omjer kompresije (max)
 - Sličnost s originalnim modelom (max)
 - NSGA-II

Module-wise similarity



- Forward pass za 64 nasumično odabrane rečenice iz wikipedije:
- Sličnost modula pažnje
- Sličnost Feed-Forward mreža
- Sličnost skrivenih stanja



*Slike preuzete iz prezentcije iz kolegija Optimiranje evolucijskim računanjem

$$[(b_0, e_0, a_0), (b_1, e_1, a_1), \dots, (b_{L-1}, e_{L-1}, a_{L-1})]$$

$$b_i, e_i \in [0, L - 1]$$

$$a_i \in \{0, 1\}$$

- Redundantncija genotipa → Fitness caching

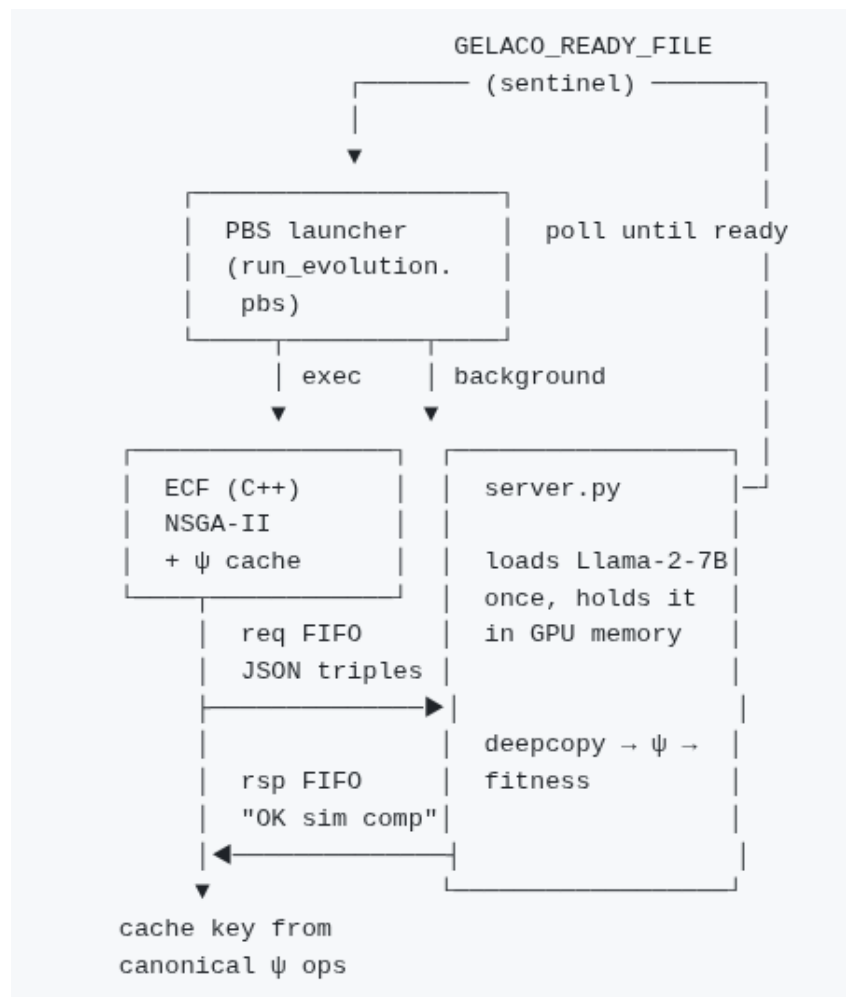
Implementacijski detalji (1)

- Llama-2-7B (L = 32)
- ECF
- Huggingface transformers
- Supek HPC klaster (A100 40 GB)
- 30 000 evaluacija, veličina populacije 200 → cca 26 sati

Implementacijski detalji (2)



ZEMRIS



Implementacijski detalji (3)



ZEMRIS

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  <NSGA2/>
</Algorithm>

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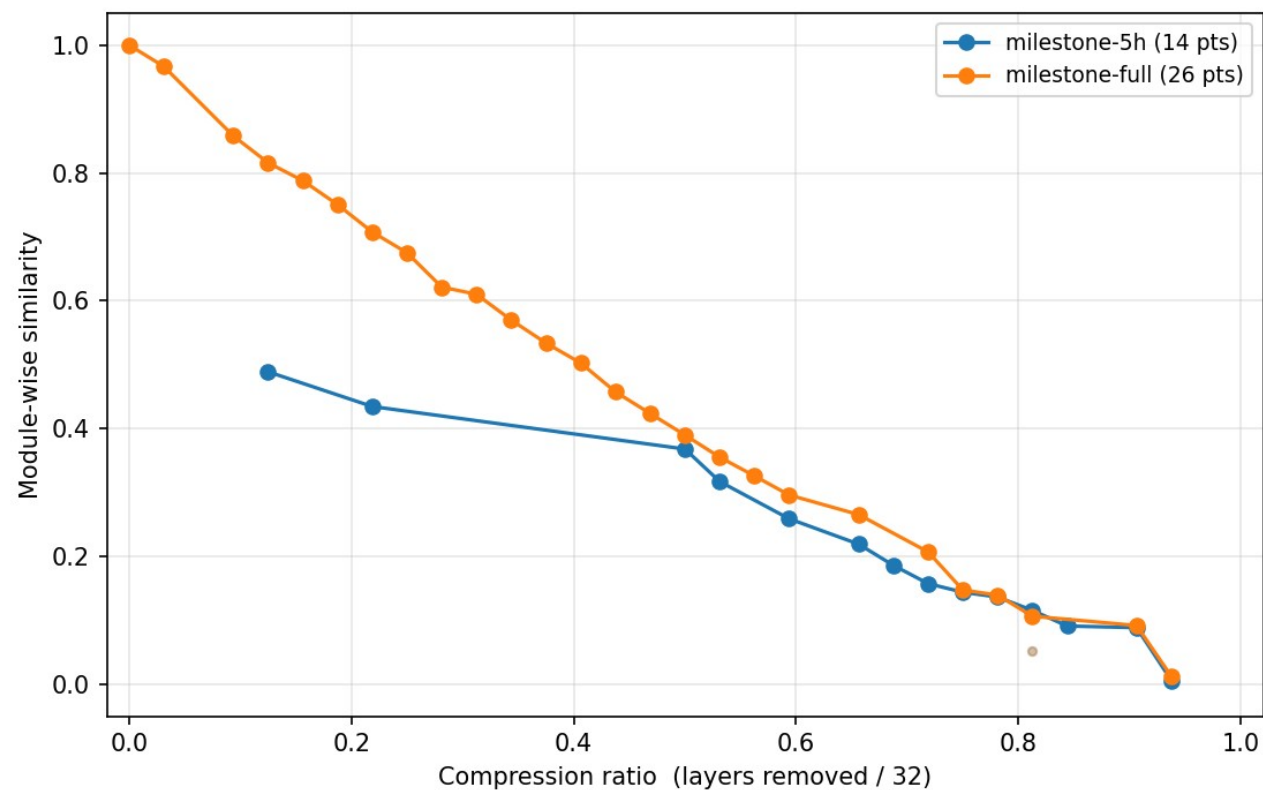
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Rezultati

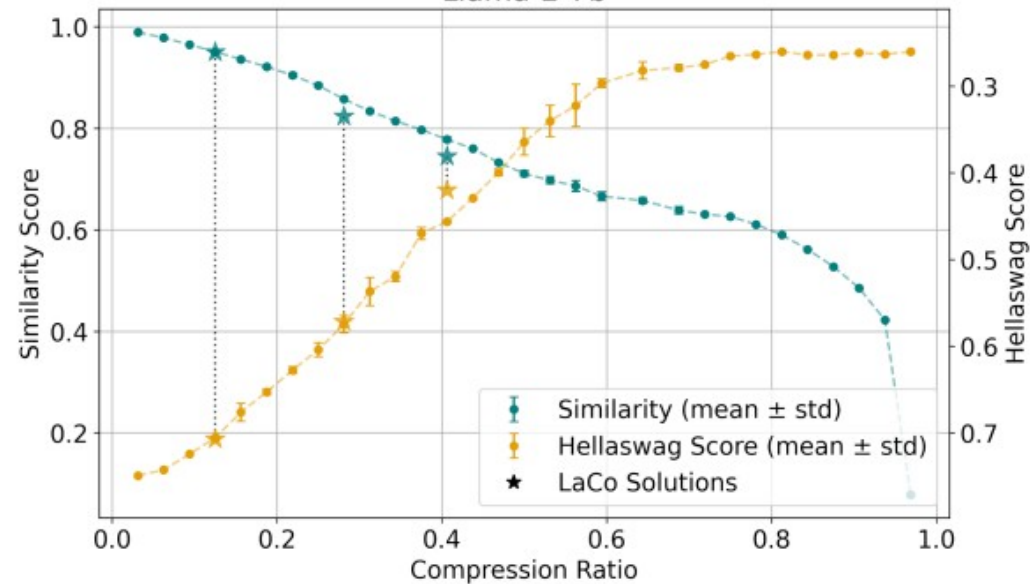


ZEMRIS

GeLaCo Pareto front



Llama-2-7b



Hvala na pozornosti!

Pitanja?

- [1] Men, X., Xu, M., Zhang, Q., Wang, B., Lin, H., Lu, Y., Han, X., Chen, W. ShortGPT: Layers in Large Language Models are More Redundant Than You Expect, 2024.
- [2] Yang, Y., Cao, Z., Zhao, H. LaCo: Large Language Model Pruning via Layer Collapse, 2024.
- [3] Ponce, D., Etchegoyhen, T., Del Ser, J. GeLaCo: An Evolutionary Approach to Layer Compression, 2025
- [4] OpenAI, Scaling Laws for Neural Language Models
- Slike koje nisu iz gornjih članaka:
 - <https://www.dl.reviews/2020/01/03/lottery-ticket-hypothesis/>
 - https://github.com/Guru-Deep-Singh/Llama-2/blob/main/images/llama_architecture.png